The Demon Wordle Problem

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3 Main Parts:

1. Wordle game base
2. Demon
3. Understanding & Defining Scrambility

Wordle, but on every new input, the target word changes, using whatever letters they got correct in position, the new word MUST have them at that location if they were guessed in the correct spot in the previous word. If letters were guessed correctly but not in the correct position, the new word MUST contain those letters, and the letters that were incorrect, the new word CANNOT contain these letters.

The new generated word must be the most challenging/hardest word, to get to from the previous target word. This is defined by the word that has the most scramblity with the correct letters. Using a **HashMap**, you can find all the words that are generatable from the word that was previous, and the word with the MOST possible scrambles with the correct parameters.

Use the dictionary to load all 5 letter words, to create the list for the wordle. You need to generate a new word, this can be a randomly generated word that MUST exist from 5 letter words in the dictionary. When you print the result, depending on what is correct or incorrect you need to color that specific letter, green for correct placement, yellow for it exists, and black for incorrect.

DEFINING SCRAMABILITY:

Scramability: How many possible permutations can be formed by the letters that are used in a selected word, excluding required letters. Calculation of requirement to make a valid permutation used:

wordlength - the length of the guess words & solve words

requiredletters - total number of correctly placed letters ("green" letters) + the incorrectly placed, correctly CONTAINED letters, (ie "yellow" letters), which will never be greater than **wordlength**

if (wordlength / requiredletters ≤ wordlength/2)

Number of letters to create a valid permutation:
(wordlength - requiredletters - 1)

OR

if (wordlength/2 ≥ wordlength/requiredletters)

Number of letters to create a valid permutation:
(wordlength - requiredletters + 1)